Hahn–Banach Norm-Preserving Extension

An End-to-End Meta-Math Walkthrough

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Abstract

The key finite step extends a dominated functional by one direction. A chain-union argument then extends it to the entire space without claiming countably many directions suffice. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

FA-05 / Functional Analysis and Measure Theory / Report v1

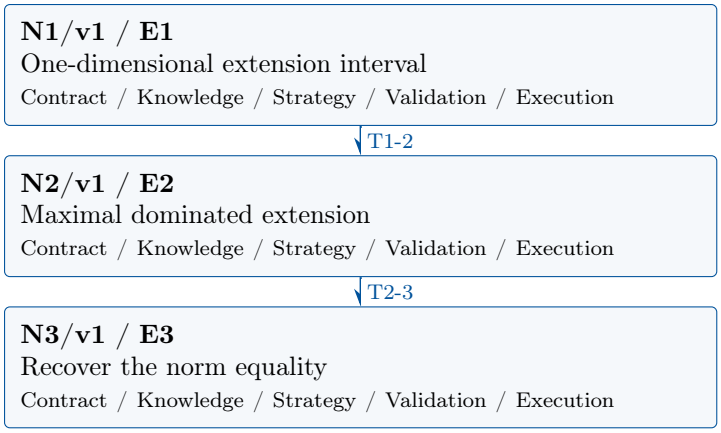
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let X be a real normed space, $M \subset X$ any linear subspace, and $f : M \to \mathbb{R}$ bounded linear. There exists bounded linear $F : X \to \mathbb{R}$ with $F|_M = f$ and $\|F\| = \|f\|$. Neither completeness of X nor closedness of M is assumed.*

A. Foundations and permitted imports

Work in set theory with Zorn's lemma: a nonempty partially ordered set in which every chain has an upper bound has a maximal element. Let $C = \|f\|$ and $p(x) = C\|x\|$, a finite-valued sublinear function: $p(x + y) \leq p(x) + p(y)$ and $p(tx) = tp(x)$ for $t \geq 0$. Use completeness of the real order to select a number between a bounded supremum and infimum.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. g extends linearly to $D + \mathbb{R}z$ while remaining dominated by p.

Receiving activity. N2.2

Compatibility. E1/v1 and all extensions of f dominated by p. Check: A union is a function only because the extensions agree on overlaps; a maximal proper domain can still be extended by E1.

T2-3: N2/E2/v1 to N3. There is a linear $F : X \to \mathbb{R}$ extending f with $F \leq p$.

Receiving activity. N3.1

Compatibility. E2/v1 and $p(x) = C\|x\|$. Check: One-sided domination alone must be converted to an absolute bound; the zero-functional case is included.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: One-dimensional extension interval

Object and role. Turn domination of $g(d) + ta$ into compatible lower and upper bounds for the new value $a = G(z)$. **Local obligation:** g extends linearly to $D + \mathbb{R}z$ while remaining dominated by p.

Inputs. A linear $g : D \to \mathbb{R}$ with $g \le p$ on a subspace D, and $z \notin D$.

C — Contract

Publish E1/v1 only after establishing this obligation: g extends linearly to $D+\mathbb{R}z$ while remaining dominated by p.

K — Knowledge

A linear $g : D \to \mathbb{R}$ with $g \le p$ on a subspace D, and $z \notin D$. Retain the stated assumptions and import versions.

S — Strategy

Turn domination of $g(d)+ta$ into compatible lower and upper bounds for the new value $a = G(z)$.

V — Validation

Both signs of t must be checked, and the choice interval must be nonempty and finite.

E — Execution

Carry out activities N1.1–N1.3 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 For $u, v \in D$, linearity and domination give $g(u) + g(v) = g(u + v) \le p(u + v) \le p(u - z) + p(v + z)$. Therefore $g(u)-p(u-z) \le p(v+z)-g(v)$. Set $L = \sup_u[g(u)-p(u-z)]$ and $U = \inf_v[p(v+z)-g(v)]$. The preceding inequality gives $L \le U$; choosing $u = 0$ and $v = 0$ bounds both endpoints between finite numbers.

N1.2 Choose $a \in [L, U]$ and define $G(d + tz) = g(d) + ta$. The representation is unique because $z \notin D$, so G is well defined and linear. For $t > 0$, $a \le p(d/t + z) - g(d/t)$ implies $G(d + tz) \le p(d + tz)$.

N1.3 For $t = -s < 0$, $a \ge g(d/s) - p(d/s - z)$ gives $g(d) - sa \le sp(d/s - z) = p(d - sz)$. The case $t = 0$ uses $g \le p$. Thus domination holds for every real scalar.

Export E1/v1. g extends linearly to $D + \mathbb{R}z$ while remaining dominated by p.

Acceptance gate. Both signs of t must be checked, and the choice interval must be nonempty and finite.

Targeted repair. If only positive scalars were considered, restore the lower endpoint L and the negative-scalar calculation before extending the domain. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Maximal dominated extension

Object and role. Order compatible extensions by domain inclusion and take unions along chains.
Local obligation: There is a linear $F : X \to \mathbb{R}$ extending f with $F \leq p$.

Inputs. E1/v1 and all extensions of f dominated by p. Incoming proof exports: E1/v1 at N2.2.

C — Contract

Publish E2/v1 only after establishing this obligation: There is a linear $F : X \to \mathbb{R}$ extending f with $F \leq p$.

K — Knowledge

E1/v1 and all extensions of f dominated by p. Incoming proof exports: E1/v1 at N2.2. Retain the stated assumptions and import versions.

S — Strategy

Order compatible extensions by domain inclusion and take unions along chains.

V — Validation

A union is a function only because the extensions agree on overlaps; a maximal proper domain can still be extended by E1.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 The set of pairs (D, g) with $M \subset D \subset X$, $g|_M = f$, and $g \leq p$ is nonempty. Order it by extension. For a chain, the union of its domains is a subspace: any two vectors and their values occur together in one chain member. The union map is consequently well defined, linear, and dominated by p.

N2.2 This union is an upper bound, so Zorn supplies a maximal pair (D, g). If $D \neq X$, choose $z \in X \setminus D$ and apply E1, contradicting maximality. Thus $D = X$; denote the resulting map by F.

Export E2/v1. There is a linear $F : X \to \mathbb{R}$ extending f with $F \leq p$.

Acceptance gate. A union is a function only because the extensions agree on overlaps; a maximal proper domain can still be extended by E1.

Targeted repair. If a union of incomparable extensions is used, restrict to chains before invoking Zorn; arbitrary extensions can assign conflicting values. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Recover the norm equality

Object and role. Apply the one-sided domination to x and $-x$, then compare with the restriction.
Local obligation: $F|_M = f$ and $\|F\| = \|f\|$.

Inputs. E2/v1 and $p(x) = C\|x\|$. Incoming proof exports: E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: $F|_M = f$ and $\|F\| = \|f\|$.

K — Knowledge

E2/v1 and $p(x) = C\|x\|$. Incoming proof exports: E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Apply the one-sided domination to x and $-x$, then compare with the restriction.

V — Validation

One-sided domination alone must be converted to an absolute bound; the zero-functional case is included.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 For every x, $F(x) \le C\|x\|$ and $-F(x) = F(-x) \le C\|x\|$. Hence $|F(x)| \le C\|x\|$, so F is bounded and $\|F\| \le C$.

N3.2 Restricting F to the unit ball of M gives $\|f\| \le \|F\|$. Since $C = \|f\|$, equality follows. If $C = 0$, this argument yields $F = 0$ and still applies.

Export E3/v1. $F|_M = f$ and $\|F\| = \|f\|$.

Acceptance gate. One-sided domination alone must be converted to an absolute bound; the zero-functional case is included.

Targeted repair. If the norm conclusion is inferred directly from $F \le p$, add the $-x$ bound and the restriction inequality. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — One-dimensional extension interval

1. **N1.1.** For $u, v \in D$, linearity and domination give $g(u)+g(v) = g(u+v) \le p(u+v) \le p(u-z)+p(v+z)$. Therefore $g(u)-p(u-z) \le p(v+z)-g(v)$. Set $L = \sup_u[g(u)-p(u-z)]$ and $U = \inf_v[p(v+z)-g(v)]$. The preceding inequality gives $L \le U$; choosing $u = 0$ and $v = 0$ bounds both endpoints between finite numbers.

2. **N1.2.** Choose $a \in [L, U]$ and define $G(d+tz) = g(d)+ta$. The representation is unique because $z \notin D$, so G is well defined and linear. For $t > 0$, $a \le p(d/t+z)-g(d/t)$ implies $G(d+tz) \le p(d+tz)$.

3. **N1.3.** For $t = -s < 0$, $a \ge g(d/s)-p(d/s-z)$ gives $g(d)-sa \le sp(d/s-z) = p(d-sz)$. The case $t = 0$ uses $g \le p$. Thus domination holds for every real scalar.

B. N2 — Maximal dominated extension

1. **N2.1.** The set of pairs (D, g) with $M \subset D \subset X$, $g|_M = f$, and $g \le p$ is nonempty. Order it by extension. For a chain, the union of its domains is a subspace: any two vectors and their values occur together in one chain member. The union map is consequently well defined, linear, and dominated by p.

2. **N2.2.** This union is an upper bound, so Zorn supplies a maximal pair (D, g). If $D \ne X$, choose $z \in X \setminus D$ and apply E1, contradicting maximality. Thus $D = X$; denote the resulting map by F.

C. N3 — Recover the norm equality

1. **N3.1.** For every x, $F(x) \le C\|x\|$ and $-F(x) = F(-x) \le C\|x\|$. Hence $|F(x)| \le C\|x\|$, so F is bounded and $\|F\| \le C$.

2. **N3.2.** Restricting F to the unit ball of M gives $\|f\| \le \|F\|$. Since $C = \|f\|$, equality follows. If $C = 0$, this argument yields $F = 0$ and still applies.

□

V. WORKED ORCHESTRATION

In $\mathbb{R}^2$ with the Euclidean norm, take $M = \{(t, t)\}$ and $f(t, t) = 2t$. The norm on M is $\sqrt{2}$. The extension $F(x, y) = x + y$ has norm $\sqrt{2}$ and restricts correctly. This is a concrete certificate; the general construction does not promise a unique extension.

A. Boundary and hypothesis audit

Uniqueness is false. In $\mathbb{R}^2$ with the ℓ^1 norm, let M be the first coordinate axis and $f(x,0) = x$. Every $F_a(x,y) = x + ay$ with $|a| \le 1$ extends f and has norm one.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N1 candidate: choose any a below the upper bound and ignore the lower bound. For $X = \mathbb{R}$, $D = \{0\}$, $p(t) = |t|$, choosing $a = -2$ satisfies $a \le 1$ but fails domination at -1, where $G(-1) = 2 > 1$. **Repair:** require $a \in [-1, 1]$, the actual two-sided interval. This corrects the extension argument under the same theorem. Recheck the maximal-extension and norm exports.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Deduce that every nonzero vector has a norming continuous linear functional by first defining it on its one-dimensional span.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. MIT OpenCourseWare, Introduction to Functional Analysis: lecture notes. See the lectures on integration, Hilbert spaces, Baire category, and Hahn–Banach.